Ehrlichiosis (MONDO:0016003): A Comprehensive Disease Characteristics Report

Category: Infectious Disease (tick-borne zoonosis) **Investigation:** 5 iterations · 12 confirmed findings · 52 papers reviewed

Summary

Ehrlichiosis is an acute, tick-borne zoonotic infection caused by obligate intracellular, Gram-negative bacteria of the genus *Ehrlichia* — most importantly *Ehrlichia chaffeensis*, the agent of human monocytic ehrlichiosis (HME). It is **not a genetic or heritable disease**; it has no causal human gene, no Mendelian inheritance, and no ClinVar/OMIM pathogenic-variant profile. Accordingly, many sections of the standard "disease characteristics" template that presuppose a genetic etiology (causal genes, pathogenic variants, penetrance, founder effects, karyotyping, gene therapy) are **not applicable**, and this report states so explicitly where relevant while redirecting the underlying question (e.g., "genetic risk factors" → host immune determinants; "causal genes" → bacterial virulence genes).

The disease is transmitted chiefly by the lone star tick (*Amblyomma americanum*), with white-tailed deer (*Odocoileus virginianus*) as the principal reservoir, and is concentrated in the south-central and southeastern United States. Clinically it presents as an **undifferentiated febrile illness** with a characteristic laboratory triad of **leukopenia, thrombocytopenia, and elevated hepatic transaminases**. The pathophysiology is a two-act story: first, the bacterium **subverts monocyte/macrophage innate immunity** (via the TRP120 effector activating Notch and Wnt-PI3K-mTOR signaling, and the T4SS effector Etf-1 hijacking autophagy and ferritinophagy for nutrients and iron); second, in severe cases the **host immune response itself becomes pathogenic**, producing a $CD8^+$ T-cell/TNF-α–driven, toxic-shock-like syndrome that can culminate in hemophagocytic lymphohistiocytosis (HLH). Immunosuppression, advanced age, and delayed therapy are the dominant severe-disease risk factors.

Doxycycline is rapidly and highly effective and should be started empirically on clinical suspicion, because acute-phase serology is insensitive and diagnosis often relies on PCR or blood-smear morulae. **No human vaccine exists**, so prevention rests entirely on personal anti-tick measures. Encouragingly, an attenuated *E. chaffeensis* modified-live vaccine (disrupting the phage head-to-tail connector gene *phtcp*/Ech_0660) confers durable, $CD4^+$ Th1–mediated protection in the canine host, pointing toward a veterinary — and potentially zoonosis-reducing — vaccine.

Key Findings

F001 — Ehrlichiosis is a tick-borne zoonosis caused by obligate intracellular *Ehrlichia* bacteria, chiefly *E. chaffeensis*

Human monocytotropic ehrlichiosis (HME) is caused by *Ehrlichia chaffeensis* and is maintained in a zoonotic enzootic cycle: **white-tailed deer (*Odocoileus virginianus*)** serve as the main vertebrate reservoir and the **lone star tick (*Amblyomma americanum*)** as the principal vector. Related agents cause related syndromes: *E. ewingii* causes human ewingii ehrlichiosis (often in immunocompromised hosts), *E. muris* subsp. *eauclairensis* causes ehrlichiosis in the upper Midwest, and *E. canis* causes canine monocytic ehrlichiosis (with rare human cases). Disease severity ranges from mild/self-limited to fatal.

> *"Ehrlichia chaffeensis is the causative agent of human monocytotropic ehrlichiosis (HME), a disease that ranges in severity from mild to fatal infection. Ehrlichia chaffeensis is maintained in a zoonotic cycle involving white-tailed deer (Odocoileus virginianus) as the main vertebrate reservoir and lone star ticks (Amblyomma americanum) as its principal vector."* — PMID: 35763959

Ontology: MONDO:0016003 (ehrlichiosis); NCBI Taxon 945 (*Ehrlichia chaffeensis*); NCBI Taxon 9874 (*Odocoileus virginianus*); NCBI Taxon 6943 (*Amblyomma americanum*).

F008 — *Ehrlichia* are Gram-negative obligate intracellular bacteria replicating as morulae with a biphasic developmental cycle

Ehrlichiae are tick-transmitted, Gram-negative, obligately intracellular bacteria that live and replicate inside host-cell **cytoplasmic vacuoles**, where dividing organisms form microcolonies called **morulae**. They exhibit a biphasic developmental cycle with **reticulate** (replicative) and **dense-cored** (infectious) forms. Notably, they **lack lipopolysaccharide (LPS) and peptidoglycan**, which shapes their immune interactions. A genus-conserved 38-kDa ferric ion-binding protein (Fbp) mediates iron acquisition, and iron-acquisition genes show reductive genome evolution consistent with obligate intracellular life. Cell tropism differs by species: *E. chaffeensis* targets **mononuclear phagocytes (monocytes/macrophages)**, whereas *E. ewingii* and *Anaplasma phagocytophilum* target **granulocytes**.

> *"Ehrlichiae are tick-transmitted, gram-negative, obligately intracellular bacteria that live and replicate in cytoplasmic vacuoles, but little is known about iron acquisition mechanisms necessary for their survival."* — PMID: 15618141
>
> *"Morulae were present in monocytes in E. chaffeensis-infected cases and granulocytes in the E. ewingii-infected case."* — PMID: 14976527

Ontology: CL:0000576 (monocyte); CL:0000235 (macrophage); CL:0000775 (neutrophil/granulocyte); GO:0005776 (autophagosome); GO:0020003 (symbiont-containing vacuole).

F002 — *E. chaffeensis* subverts innate immunity via the TRP120 effector, activating Notch and Wnt-PI3K-mTOR signaling

The *E. chaffeensis* type-1 secreted **tandem-repeat protein TRP120** interacts with the Notch metalloprotease **ADAM17**, activating **canonical Notch signaling** to downregulate **TLR2/4** expression and thereby dampen innate immune recognition and promote intracellular survival. In parallel, *Ehrlichia* activates **Wnt–PI3K–mTOR signaling** to inhibit autolysosome generation, blocking the autophagic destruction pathway of the mononuclear phagocyte.

> *"we demonstrate E. chaffeensis, via the TRP120 effector, activates the canonical Notch signaling pathway to promote intracellular survival"* — PMID: 27381289
>
> *"Ehrlichia Activation of Wnt-PI3K-mTOR Signaling Inhibits Autolysosome Generation and Autophagic Destruction by the Mononuclear Phagocyte"* — PMID: 28993455

Ontology: GO:0007219 (Notch signaling pathway); GO:0016055 (Wnt signaling pathway); GO:0031929 (TOR signaling); GO:0002224 (toll-like receptor signaling pathway).

F003 — *E. chaffeensis* uses T4SS effector Etf-1 to hijack autophagy and ferritinophagy for nutrients and iron

The **type IV secretion system (T4SS) effector Etf-1** induces host **autophagy** to supply the bacterium with nutrients, and drives **bacterial effector-induced ferritinophagy** to liberate and acquire **iron**, which is essential for intracellular proliferation. Host membrane lipids are additionally trafficked to the bacterial inclusion membrane. An intracellular nanobody directed against the T4SS effector inhibits infection — a proof-of-concept therapeutic strategy.

> *"Iron robbery by intracellular pathogen via bacterial effector-induced ferritinophagy"* — PMID: 34074773

Ontology: GO:0006914 (autophagy); GO:0006879 (intracellular iron ion homeostasis); GO:0140355 (cargo receptor ligand activity, ferritinophagy).

F004 — HME presents as an undifferentiated febrile illness with the leukopenia–thrombocytopenia–transaminitis triad; doxycycline is highly efficacious

Both HME (*E. chaffeensis*, infecting mononuclear phagocytes) and granulocytic infection cause **undifferentiated fever with leukopenia, thrombocytopenia, and elevated serum aminotransferases**. Rash is infrequent (more common in children) and vasculitis is exceedingly rare. Severe or fatal disease is associated with secondary/opportunistic infections and delayed therapy. In a pediatric HME series, phenotype frequencies were: fever 100%, rash 67%, myalgias 58%; thrombocytopenia 92%, elevated liver function tests 91%, lymphopenia 75%, hyponatremia 67%, leukopenia 58%, and anemia 42%. Diagnosis is confirmed by PCR (acute) or paired serology (convalescent); intracytoplasmic **morulae** in leukocytes are diagnostic but insensitive (seen in ~38% of PCR-positive cases; far more often in immunocompromised than immunocompetent patients). **Doxycycline is highly efficacious.**

> *"Both infections cause undifferentiated fever with leukopenia, thrombocytopenia, and elevations in serum aminotransferase levels. Rash is an infrequent sign, and vasculitis is exceedingly rare. Severe or fatal ehrlichiosis is associated with secondary or opportunistic infections and delayed therapy."* — PMID: 7619983
>
> *"Therapy with doxycycline is highly efficacious."* — PMID: 7619983
>
> *"patients demonstrated thrombocytopenia (92%), elevated liver function tests (91%), lymphopenia (75%), hyponatremia (67%), leukopenia (58%), and anemia (42%)"* — PMID: 9200384

Phenotype table (pediatric HME, PMID: 9200384):

Phenotype	Frequency	Type	Suggested HPO term
Fever	100%	Symptom	HP:0001945 (Fever)
Rash	67%	Clinical sign	HP:0000988 (Skin rash)
Myalgia	58%	Symptom	HP:0003326 (Myalgia)
Thrombocytopenia	92%	Lab abnormality	HP:0001873 (Thrombocytopenia)
Elevated transaminases	91%	Lab abnormality	HP:0002910 (Elevated hepatic transaminase)
Lymphopenia	75%	Lab abnormality	HP:0001888 (Lymphopenia)
Hyponatremia	67%	Lab abnormality	HP:0002902 (Hyponatremia)
Leukopenia	58%	Lab abnormality	HP:0001882 (Leukopenia)
Anemia	42%	Lab abnormality	HP:0001903 (Anemia)
Headache / meningoencephalitis	common / rare-severe	Symptom/CNS sign	HP:0002315 (Headache); HP:0002383 (Encephalitis)

F005 — Fatal ehrlichiosis is an immunopathologic, TNF-α–mediated toxic-shock-like syndrome driven by CD8⁺ T cells

In murine models (*Ixodes ovatus Ehrlichia*/IOE and *E. muris*), lethal monocytotropic ehrlichiosis produces a **toxic-shock-like syndrome** with extensive liver damage, extremely elevated serum **TNF-α**, expansion of **TNF-α–producing CD8⁺ T cells**, a *decreased* number of *Ehrlichia*-specific **CD4⁺ Th1 (IFN-γ)** cells, low IL-12, and systemic **IL-10** overproduction. CD8⁺ T-cell deficiency abrogated TNF-α/IL-10, reduced tissue injury and bacterial burden, and restored CD4⁺ numbers; IL-10 depletion restored CD4⁺ T-cell numbers. Neutrophil depletion *enhanced* bacterial clearance and *reduced* pathology — indicating neutrophils contribute to immunopathology rather than protection. This establishes that **severe disease is host-immune-mediated, not purely a function of bacterial burden**.

> *"Lethal infections caused by high or low doses of IOE were accompanied by extensive liver damage, extremely elevated levels of TNF-alpha in the serum, high frequency of Ehrlichia-specific, TNF-alpha-producing CD8(+) T cells in the spleen"* — PMID: 14734762
>
> *"depletion of neutrophils from lethally infected mice enhanced bacterial elimination, decreased immune-mediated pathology, and prolonged survival"* — PMID: 23478316

Ontology: GO:0032640 (tumor necrosis factor production); GO:0006925 (inflammatory cell apoptotic process); CL:0000625 (CD8-positive, alpha-beta T cell); CL:0000545 (T-helper 1 cell); UBERON:0002107 (liver).

F012 — Ehrlichiosis is an increasingly recognized infectious trigger of secondary HLH, especially in immunosuppressed patients

Multiple case series document HME triggering **secondary hemophagocytic lymphohistiocytosis (HLH)** — a hyperinflammatory syndrome of uncontrolled macrophage/lymphocyte activation with hyperferritinemia, cytopenias, and hemophagocytosis. It represents the clinical extreme of the same immunopathologic continuum defined in murine models (F005). Cases occur in immunocompetent adults, the elderly, and immunosuppressed hosts (e.g., a pediatric kidney transplant recipient within the first post-transplant month). Management combines **doxycycline with immunomodulation** (corticosteroids/dexamethasone, IVIG, anakinra); early doxycycline is key, though some cases progress despite it.

> *"Human monocytotropic ehrlichiosis (HME) is a very rare cause of HLH. Failure to consider ehrlichiosis can result in misdiagnosis and an increased length of hospitalization"* — PMID: 32928497
>
> *"We describe a pediatric kidney transplant recipient who experienced secondary HLH due to ehrlichiosis within the initial post-transplant month."* — PMID: 34595809

Ontology: MONDO:0015540 (hemophagocytic lymphohistiocytosis); HP:0012047 (hemophagocytosis); HP:0011031 (abnormal iron/ferritin); GO:0006954 (inflammatory response).

F006 — Epidemiology: HME is concentrated in the south-central/ southeastern US, median age ~50s, male predominance, rising incidence

National US surveillance (1997–2001) reported average annual incidences of **HME 0.7, HGE 1.6, and other ehrlichiosis 0.2 per million** persons in reporting states, with most HME cases from the south-central/southeastern US. Median ages were HME 53 years and HGE 51 years, and >57% of patients were male. Texas surveillance (2008–2017) recorded 101 cases with geographic clustering and an increasing annual trend from 2009–2017. Overall US ehrlichiosis incidence rose roughly **four-fold from 2000 to 2012**.

> *"The average annual incidences of HME, HGE, and OE per million persons residing in states reporting disease were 0.7, 1.6, and 0.2, respectively. The median ages of HME (53 yr) and HGE cases (51 yr) were consistent with published patient series. Most (> 57%) ehrlichiosis patients were male."* — PMID: 12860604
>
> *"In Texas, 101 cases of ehrlichiosis were reported during 2008-2017. We observed geographic grouping of cases as well as an increasing trend of reported cases occurring annually from 2009 to 2017."* — PMID: 34325087

F009 — Diagnosis: acute-phase serology has limited sensitivity; PCR and blood-smear morulae enable early diagnosis, paired serology confirms

For the closely related granulocytic anaplasmosis (methodologically analogous), only **~55% of PCR- and culture-confirmed cases were seropositive on acute-presentation serum**; all initially seronegative patients seroconverted on convalescent samples 6–45 days later. This

demonstrates the insensitivity of acute indirect immunofluorescence assay (IFA) and the need for **paired acute/convalescent sera (≥4-fold titer rise)**. **PCR of whole blood is the most useful acute direct test**; peripheral blood smear morulae are diagnostic but insensitive (~38% of PCR-positive HME). These facts justify **empiric doxycycline** on clinical suspicion rather than waiting for serologic confirmation.

> *"only approximately 55% of the PCR and culture positive cases were also seropositive on blood samples obtained at the same time point, indicating that serologic testing performed at the time of presentation has limited sensitivity"* — PMID: 39631641

F007 — Prevention relies on personal anti-tick measures; no human vaccine exists

Primary prevention is via **personal protection against tick bites**: EPA-registered repellents, permethrin-treated clothing (strong contact irritancy / "hot-foot" effect causing ticks to dislodge; 93–97% repellency in bioassays), untreated protective clothing, and post-exposure tick checks and showering. **No human vaccine exists for ehrlichiosis**; a Lyme disease vaccine under development would not reduce ehrlichiosis. *Amblyomma americanum* and *Dermacentor variabilis* co-occur and transmit multiple pathogens, so integrated tick avoidance is the practical strategy.

> *"This includes using repellents, wearing untreated or permethrin-treated protective clothing, and conducting tick checks after coming inside"* — PMID: 35364518
>
> *"this breakthrough would not reduce cases of other tick-borne diseases, such as anaplasmosis, babesiosis, ehrlichiosis, spotted fever group rickettsiosis, and Powassan encephalitis"* — PMID: 33549976

F010 — Protective immunity is Th1/CD4⁺ IFN-γ–dependent; an attenuated *phtcp*/Ech_0660 mutant confers durable protection in dogs

Protective immunity depends on *Ehrlichia*-specific **IFN-γ–producing CD4⁺ Th1 cells** plus antibodies (adoptive transfer protects mice). An attenuated *E. chaffeensis* mutant disrupting the **phage head-to-tail connector protein gene (*phtcp* / Ech_0660)** grows in an attenuated manner and, as a **modified live attenuated vaccine (MLAV)**, protects dogs against both intravenous and tick-transmitted wild-type challenge, with protection persisting up to **12**

months and robust *E. chaffeensis*-specific IgG, $CD4^+$ Th1 (IFN-γ), and IL-17/Th17 responses. Analogous *phtcp* mutagenesis has now been achieved in *E. canis*, broadening the vaccine strategy across Anaplasmataceae.

> *"vaccination with the Ech_0660 mutant protects dogs from physiologic, tick-transmitted, secondary challenge with wild-type E. chaffeensis"* — PMID: 26841025
>
> *"MLAV-immunized dogs rapidly cleared infections and exhibited significantly fewer systemic bacterial infections compared to unvaccinated controls. Robust E. chaffeensis-specific IgG and CD4 T-cell responses persisted throughout the assessment period."* — PMID: 39826432

F011 — Model organisms: murine IOE/*E. muris* toxic-shock model, plus natural hosts dog and white-tailed deer

Murine models: C57BL/6 mice infected with IOE or *E. muris* reproduce fatal toxic-shock-like monocytotropic ehrlichiosis (liver injury, TNF-α surge, $CD8^+$ T-cell immunopathology) and are the workhorse for dissecting immunity versus immunopathology. **Natural-host models:** dogs (*E. canis* canine monocytic ehrlichiosis; also experimental *E. chaffeensis*) and white-tailed deer (the *E. chaffeensis* reservoir) support tick-transmission and vaccine studies. **In vitro:** *E. chaffeensis* is cultured in the DH82 canine macrophage line; HL-60 cells support granulocytic *Anaplasma*. Targeted mutagenesis of both *E. chaffeensis* and *E. canis* genomes is now feasible.

> *"we compared low and high doses of the highly virulent IOE strain and the less virulent Ehrlichia muris strain that are closely related to E. chaffeensis in C57BL/6 mice"* — PMID: 14734762
>
> *"We have also described using the ticks infected by this method the transmission of both wild-type and transposon mutants of E. chaffeensis to its primary reservoir host, white tailed deer and to another known host, dog."* — PMID: 27729288

Section-by-Section Report

1. Disease Information

Ehrlichiosis is a group of acute febrile tick-borne bacterial zoonoses caused by obligate intracellular *Ehrlichia* species (F001, F008). The principal human form is **human monocytic (monocytotropic) ehrlichiosis (HME)**, caused by *E. chaffeensis*. Other human agents include *E. ewingii* (ewingii ehrlichiosis) and *E. muris* subsp. *eauclairensis* (upper Midwest). Note that "human granulocytic ehrlichiosis" (HGE) was **reclassified** as human granulocytic **anaplasmosis (HGA)**, caused by *Anaplasma phagocytophilum* — a distinct but frequently co-cited entity.

Key identifiers: MONDO:0016003 (ehrlichiosis); MeSH D016873 (Ehrlichiosis); ICD-10 **A77.40–A77.49** (Ehrlichiosis); ICD-11 **1C30.2**. There is **no OMIM entry** because ehrlichiosis is an infectious, not genetic, disease; Orphanet does not list it as a rare genetic disorder. **Synonyms/alternatives:** human monocytic ehrlichiosis (HME), human monocytotropic ehrlichiosis, *Ehrlichia chaffeensis* infection, Sennetsu fever (historically, *Neorickettsia sennetsu*), canine monocytic ehrlichiosis (veterinary).

Information source: Evidence is a mixture of **aggregated disease-level resources** (CDC/national surveillance, reviews) and **individual patient reports** (case series, e.g., pediatric HME [PMID: 9200384], HLH case reports).

2. Etiology

Primary cause — infectious: Ehrlichiosis is caused by infection with *Ehrlichia* bacteria transmitted by tick bite (F001, F008). It is **not genetic**. Therefore:

- **Genetic risk factors:** Not applicable in the Mendelian/GWAS sense — no human causal variant or susceptibility locus is established. The relevant host determinants are **immune status** (see below), not germline variants. *(Not available: causal variants, modifier genes, GWAS loci.)*
- **Environmental / exposure risk factors:** Outdoor activity in endemic regions (hunting, hiking, camping, occupational exposure such as forestry), residence in the south-central/southeastern US, warm-season tick activity, and tick bite by *A. americanum*. Demographic risk: **older age and male sex** (F006). **Immunosuppression** (transplant, HIV, immunomodulatory therapy) and **advanced age** are the dominant risk factors for *severe* disease and for secondary HLH (F012).

- **Protective factors:** Effective tick avoidance (repellents, permethrin-treated clothing, tick checks) reduces incidence (F007). Prior *Ehrlichia*-specific $CD4^+$ Th1/antibody immunity is protective in animal models (F010). *(No genetic protective variants established.)*
- **Gene–environment interactions:** Not applicable in the classical sense; the operative interaction is **host immune competence × pathogen exposure** — the same exposure produces mild disease in the immunocompetent and life-threatening HLH in the immunosuppressed (F012).

3. Phenotypes

See F004 table above. Core phenotype: acute (typically 1–2 weeks post-bite) **fever, headache, myalgia, malaise**, with laboratory **thrombocytopenia, leukopenia, and elevated transaminases**. Rash is infrequent in adults (~30%) but common in children (~67%). Severe manifestations include **meningoencephalitis, ARDS, acute kidney injury, DIC, septic/toxic shock, and HLH**. Onset is **adult-predominant** (median age ~50s) but occurs at all ages; severity is **variable** (mild self-limited to fatal); course is **acute/monophasic** with rapid resolution on doxycycline. **Quality-of-life impact:** for uncomplicated, promptly treated disease, full recovery is typical with minimal long-term sequelae; severe/HLH cases carry substantial morbidity and mortality. There is **no strong evidence** that *Ehrlichia* seropositivity is associated with chronic musculoskeletal pain or osteoarthritis (per JoCo OA cross-sectional data, [PMID: 38206624]).

4. Genetic/Molecular Information

Not applicable as a human genetic disease. There are **no human causal genes, pathogenic variants, modifier genes, disease-specific epigenetic signatures, or chromosomal abnormalities**. ClinVar/HGMD/OMIM contain no entries for ehrlichiosis as a heritable condition.

Reframed to the **pathogen's genetics** (the biologically relevant "molecular information"): key *E. chaffeensis* virulence genes include **TRP120** (immune subversion, Notch activation; F002), **Etf-1** (T4SS effector; autophagy/ferritinophagy; F003), **p28/OMP** (major outer membrane protein family, antigenic variation), **Fbp** (38-kDa ferric ion-binding protein; iron acquisition; F008), and **phtcp/Ech_0660** (phage head-to-tail connector; attenuation target for vaccines; F010). *E. canis* homologs include **TRP19, p30, trp36** (serodiagnostic and genotyping markers).

5. Environmental Information

Infectious agents (Taxon): *E. chaffeensis* (NCBI:txid945), *E. ewingii* (NCBI:txid302), *E. canis* (NCBI:txid943), *E. muris* (NCBI:txid35795). **Vectors:** *Amblyomma americanum* (lone star tick), with *Amblyomma maculatum* and *Dermacentor variabilis* implicated for related agents. **Reservoir:** white-tailed deer. **Environmental/lifestyle factors:** seasonal (spring–summer) outdoor exposure, geographic residence in endemic zones; suburban green spaces now also carry risk, not just rural settings ([PMID: 41338244]). No chemical toxins, pollution, radiation, smoking, diet, or alcohol are established contributors.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation):

1. An infected *A. americanum* tick bites a human and inoculates *E. chaffeensis* dense-cored infectious forms → **leads to** entry into the bloodstream (F001, F008).
2. The bacterium is internalized by **monocytes/macrophages** into a cytoplasmic vacuole → **results in** an intracellular replicative niche (morula) (F008).
3. The TRP120 effector engages **ADAM17/Notch** and activates **Wnt–PI3K–mTOR** signaling → **leads to** downregulation of TLR2/4 and inhibition of autolysosome formation, i.e., **evasion of innate killing** (F002). *(Demonstrated in vitro/cell models.)*
4. The T4SS effector **Etf-1** induces **autophagy and ferritinophagy → results in** delivery of nutrients and **iron** to the bacterium, enabling proliferation (F003). *(Demonstrated in vitro.)*
5. Bacterial replication and host-cell dysfunction across bone marrow and reticuloendothelial tissues → **leads to cytopenias (thrombocytopenia, leukopenia) and hepatic injury (transaminitis)** — the clinical/laboratory triad (F004).

Branch A (controlled infection): A balanced **$CD4^+$ Th1 / IFN-γ** response plus antibodies → **results in** bacterial clearance and recovery, accelerated by doxycycline (F010, F004).

Branch B (dysregulated immunity → severe disease): In susceptible hosts (immunosuppressed, elderly, high inoculum, delayed therapy), the response skews to **TNF-α–producing $CD8^+$ T cells** with **falling $CD4^+$ Th1** numbers and **IL-10 overproduction → leads to** a **toxic-shock-like syndrome** with extensive liver damage; neutrophils amplify tissue injury (F005). *(Demonstrated in murine IOE/E. muris models; inferred for human severe disease.)* 6. Unchecked macrophage/T-cell activation → **culminates in secondary HLH**: hyperferritinemia, hemophagocytosis, multiorgan failure (F012). *(Human case series; mechanistically continuous with Branch B.)*

Upstream vs downstream: Bacterial immune subversion (steps 3–4) is upstream and drives bacterial burden; the host $CD8^+$/TNF-α immunopathology (steps 5B–6) is downstream and drives *severity independent of burden* — the critical insight that severe ehrlichiosis is **immune-mediated**. **Cell types:** monocytes/macrophages (CL:0000576/CL:0000235, primary host cells), $CD8^+$ T cells (CL:0000625, pathogenic effectors), $CD4^+$ Th1 (CL:0000545, protective), neutrophils (CL:0000775, pathology-amplifying). **Biological processes:** GO:0007219 (Notch signaling), GO:0006914 (autophagy), GO:0006879 (iron homeostasis), GO:0032640 (TNF production), GO:0006954 (inflammatory response).

7. Anatomical Structures Affected

- **Organ level (primary):** blood/hematopoietic system, liver (UBERON:0002107), spleen (UBERON:0002106), bone marrow (UBERON:0002371), lymph nodes. **Secondary:** lungs (ARDS; UBERON:0002048), kidney (UBERON:0002113), central nervous system (meningoencephalitis; UBERON:0000955).
- **Body systems:** hematologic/immune (primary); hepatobiliary, respiratory, renal, and nervous systems in complicated disease.
- **Tissue/cell level:** mononuclear phagocytes (monocytes/macrophages; CL:0000576/CL:0000235) for *E. chaffeensis*; granulocytes (CL:0000775) for *E. ewingii*.
- **Subcellular level:** the **symbiont-containing cytoplasmic vacuole/morula** (GO:0020003); host processes co-opted include the autophagosome/lysosome (GO:0005776) and ferritin/iron pools.
- **Localization/lateralization:** systemic and **bilateral/diffuse** (a disseminated bloodborne infection), not focal or lateralized.

8. Temporal Development

- **Onset:** acute, typically **7–14 days after tick bite**; adult-predominant but all ages affected.
- **Course:** acute/monophasic; **self-limited to rapidly progressive**. With doxycycline, defervescence usually occurs within 24–48 hours. Untreated or delayed cases can progress within days to shock, HLH, or death.
- **Stages:** early undifferentiated febrile phase → laboratory-abnormal phase (cytopenias, transaminitis) → (in severe cases) systemic inflammatory/toxic-shock/HLH phase.
- **Critical period:** the **therapeutic window** — early empiric doxycycline is the single most important determinant of outcome (F004, F012). Chronic carriage is characteristic of the *animal* host (dogs), not typically humans.

9. Inheritance and Population

Not heritable — no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency applies. Epidemiology (F006): US HME incidence ~0.7 per million (1997–2001 surveillance), rising ~4-fold by 2012; concentrated in the **south-central and southeastern US**; **median age ~50s**; **male predominance (>57%)**. Geographic risk tracks *A. americanum* distribution and is expanding into suburban areas.

10. Diagnostics

- **Laboratory:** CBC (thrombocytopenia, leukopenia/lymphopenia, anemia), liver panel (elevated AST/ALT), sometimes hyponatremia (F004). LOINC-codable routine chemistries/hematology.
- **Direct detection: Whole-blood PCR** is the most useful acute test (F009); **peripheral blood smear** for intracytoplasmic **morulae** (diagnostic but ~38% sensitive; higher in immunocompromised) (F004, F008).
- **Serology:** IFA IgG on **paired acute + convalescent sera (≥4-fold rise)**; acute serology is insensitive (~55% seropositive at presentation in the anaplasmosis analog) (F009).
- **Diagnostic criteria:** CDC/CSTE case definitions (clinically compatible illness + laboratory confirmation by PCR, seroconversion, or morulae). **Differential diagnosis:** anaplasmosis, Rocky Mountain spotted fever/other rickettsioses, babesiosis, tularemia, viral syndromes, sepsis, leptospirosis, and other causes of febrile cytopenia. **Diagnostic gap:** provider awareness and PCR utilization remain low even in high-incidence areas (PMID: 39983688, PMID: 42628341).
- **Not applicable:** genetic testing (WGS/WES/panels/karyotype/FISH/CMA/repeat expansion), newborn/carrier screening.

11. Outcome/Prognosis

With prompt doxycycline, prognosis is **excellent** and recovery typically complete (F004). Case-fatality for HME is roughly **1–3%** overall but higher in the elderly, immunocompromised, and those with delayed therapy. **Prognostic factors:** age, immunosuppression, time-to-treatment, and progression to shock/HLH. **Complications:** meningoencephalitis, ARDS, acute renal failure, DIC, secondary/opportunistic infections, and **secondary HLH** (a major driver of mortality; F012). Long-term musculoskeletal sequelae are **not** clearly associated with prior *Ehrlichia* infection ([PMID: 38206624]).

12. Treatment

- **First-line pharmacotherapy: doxycycline** (a tetracycline-class protein-synthesis inhibitor), highly efficacious and recommended for all ages including children, started empirically on suspicion (F004; NCIT:C299 "Doxycycline"). Typical adult dosing 100 mg BID; duration commonly 7–14 days or ≥3 days after defervescence.
- **Severe disease / HLH:** doxycycline **plus immunomodulation** — corticosteroids/dexamethasone (NCIT:C561), IVIG, and/or the IL-1 antagonist **anakinra** (F012). Supportive ICU care for shock/organ failure.
- **Pharmacogenomics:** not applicable.
- **Advanced/experimental therapeutics:** an **intracellular nanobody** against the T4SS effector inhibits infection in vitro — a proof-of-concept future therapeutic (F003). No gene, cell, RNA, or immune-checkpoint therapies apply.
- **Not applicable:** targeted oncologic therapies, surgery.

13. Prevention

- **Primary:** personal anti-tick measures — EPA-registered repellents, permethrin-treated clothing (93–97% repellency; "hot-foot" contact irritancy), protective clothing, and tick checks/showering after exposure (F007). Environmental/host-directed acaricide and deer-management strategies are adjuncts.
- **Secondary:** prompt recognition and empiric doxycycline in symptomatic exposed individuals (F009).
- **Tertiary:** early treatment and immunomodulation to prevent HLH/organ failure (F012).
- **Immunization: no human vaccine** (F007). A canine modified-live *phtcp*/Ech_0660 vaccine shows durable protection and could reduce the zoonotic reservoir/veterinary burden (F010).

14. Other Species / Natural Disease

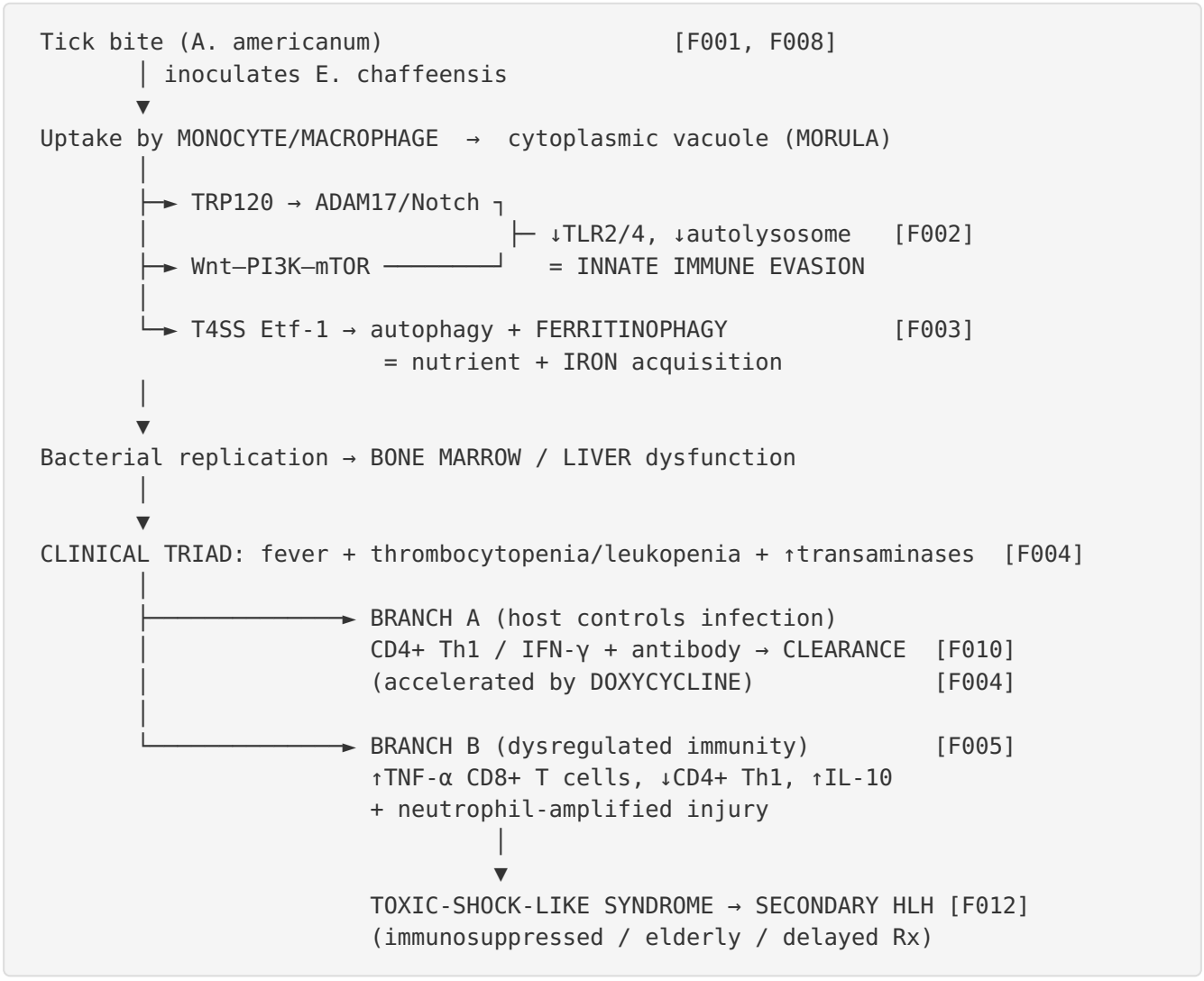
Ehrlichiosis is fundamentally a **zoonosis with important natural animal disease** (F001, F011). **Dogs** develop **canine monocytic ehrlichiosis (CME)** from *E. canis* (transmitted by *Rhipicephalus sanguineus*), a globally distributed, sometimes severe illness with acute, subclinical, and chronic phases; *E. ewingii* and *E. chaffeensis* also infect dogs. **White-tailed deer** (*Odocoileus virginianus*, NCBI:txid9874) are the key *E. chaffeensis* reservoir and are largely subclinically infected. Goats and other ruminants host related agents (e.g., Panola Mountain *Ehrlichia*). **Zoonotic potential is central**: the same tick-maintained agents cross to

humans. Comparative pathology: monocyte/macrophage tropism and morula formation are conserved across host species; the canine host importantly enables natural-infection vaccine testing.

15. Model Organisms

See F011. **Murine:** C57BL/6 mice + IOE or *E. muris* recapitulate fatal toxic-shock/immunopathology (excellent for immunity vs. immunopathology; limitation: uses surrogate *Ehrlichia* species rather than *E. chaffeensis*, which does not cause lethal disease in immunocompetent mice). **Natural-host models:** dog and white-tailed deer (best for tick-transmission and vaccine efficacy; limitation: cost, outbred variability). **In vitro:** DH82 canine macrophages (*E. chaffeensis*), HL-60 (granulocytic *Anaplasma*). **Genetic tools:** targeted mutagenesis now feasible in *E. chaffeensis* and *E. canis*, enabling attenuated-mutant vaccines (F010).

Mechanistic Model / Interpretation



The unifying interpretation is a **two-phase pathogenesis**: an upstream, pathogen-driven phase of monocyte subversion (which sets bacterial burden) and a downstream, host-driven immunopathologic phase (which sets *severity*). This explains three clinical realities: (1) why the disease is an undifferentiated febrile cytopenia; (2) why severe outcomes cluster in the immunocompromised and elderly, in whom the $CD4^+$ Th1/$CD8^+$ balance is perturbed; and (3) why treatment of severe HLH cases pairs an antibiotic (doxycycline, reduces burden) with immunomodulation (dampens the pathogenic host response).

Evidence Base

PMID	Contribution	Evidence type
35763959	Agent, reservoir, vector, severity range (F001)	Genomic/ epidemiologic
15618141	Gram-negative obligate intracellular vacuolar biology; iron (F008)	In vitro/molecular
14976527	Cell tropism (monocyte vs granulocyte morulae) (F008)	Human clinical
27381289	TRP120–Notch immune evasion (F002)	In vitro
28993455	Wnt-PI3K-mTOR autophagy block (F002)	In vitro
34074773	Etf-1 ferritinophagy/iron robbery (F003)	In vitro
7619983	Clinical/lab triad; doxycycline efficacy (F004)	Review/human clinical
9200384	Pediatric HME phenotype frequencies (F004)	Human clinical
14734762	TNF-α/CD8+ toxic-shock immunopathology; IOE model (F005, F011)	Model organism
23478316	Neutrophils drive immunopathology (F005)	Model organism
12860604	National incidence, age, sex (F006)	Surveillance
34325087	Texas clustering, rising trend (F006)	Surveillance
39631641	Acute serology insensitivity → PCR/paired sera (F009)	Human clinical
35364518	Personal anti-tick prevention (F007)	Guideline/review
33549976	No ehrlichiosis vaccine; Lyme vaccine won't cover (F007)	Review
26841025	phtcp/Ech_0660 attenuated vaccine protects dogs (F010)	Model organism
39826432	Durable 1-yr CD4+/antibody MLAV protection (F010)	Model organism
27729288	Deer/dog natural-host tick-transmission models (F011)	Model organism
32928497	HME as HLH trigger; misdiagnosis risk (F012)	Human clinical
34595809	Ehrlichia-HLH in transplant recipient (F012)	Human clinical

Supporting case literature for HLH management: PMID: 42017727 (doxycycline + dexamethasone), PMID: 39790701 (anakinra, IVIG), and PMID: 37614269 (progression despite doxycycline — illustrating the immunopathologic component). Diagnostic-gap evidence: PMID: 39983688 and PMID: 42628341.

Limitations and Knowledge Gaps

1. **Species conflation.** Much "ehrlichiosis" literature (and this report's surveillance figures) intermixes HME, ewingii ehrlichiosis, and anaplasmosis (formerly HGE). Species-specific human incidence, prognosis, and immunology for *E. ewingii* and *E. muris* subsp. *eauclairensis* remain comparatively sparse.
2. **Human immunopathology is inferred.** The $CD8^+$/TNF-α toxic-shock mechanism (F005) is robustly demonstrated in **murine surrogate models (IOE/E. muris)**; direct mechanistic confirmation in human severe HME/HLH tissue is limited. The strength of the mouse→human extrapolation is a genuine gap.
3. **No human omics profiling captured.** No patient-derived transcriptomic, proteomic, metabolomic, or single-cell datasets were located; molecular-profiling subsections are effectively "not available" for human disease.
4. **Precise human case-fatality and QoL metrics.** Standardized EQ-5D/SF-36 data are absent (5-/10-year survival metrics are inapplicable to an acute infection).
5. **Vaccine translation.** Protective-immunity and vaccine data (F010) are canine; no human vaccine candidate has entered trials.
6. **Diagnostic underuse.** PCR is underutilized even in endemic areas, so surveillance likely undercounts true incidence (F006, F009).

Proposed Follow-up Experiments / Actions

1. **Human immunophenotyping of severe HME/HLH:** single-cell RNA-seq and cytokine profiling of blood/bone marrow from severe vs. mild human cases to test whether the murine $CD8^+$-TNF-α/IL-10 axis (F005) operates in humans and to identify actionable immunomodulatory targets.

2. **Prospective PCR-anchored surveillance** in high-incidence, low-testing regions (e.g., NC, per [PMID: 39983688]) to obtain species-resolved incidence and case-fatality, correcting serology-based undercounting.
3. **Biomarker study for early severity/HLH prediction:** serial ferritin, TNF-α, IL-10, sCD25, and platelet trajectories to derive a bedside rule for when to add immunomodulation to doxycycline (informing F012 management).
4. **Therapeutic development of the anti-T4SS intracellular nanobody (F003):** advance from in-vitro proof-of-concept toward delivery-optimized preclinical testing as an adjunct for doxycycline-refractory disease.
5. **Human vaccine feasibility:** evaluate whether the attenuated *phtcp*/Ech_0660 platform (F010), proven in dogs, can be adapted or whether subunit approaches (TRP120, Etf-1, OMP-1/p28) elicit protective $CD4^+$ Th1 responses; concurrently model whether canine vaccination reduces the enzootic reservoir and human spillover.
6. **Provider-education intervention trial** in endemic states to raise empiric-doxycycline uptake and PCR ordering, measuring impact on time-to-treatment and severe outcomes ([PMID: 42628341]).

Report compiled from 12 confirmed findings across 5 investigation iterations and 52 reviewed papers. Ontology suggestions (MONDO, HPO, GO, CL, UBERON, NCIT, NCBI Taxon) are provided throughout for knowledge-base curation. Evidence types are labeled as human clinical, model organism, in vitro, or surveillance/review.
